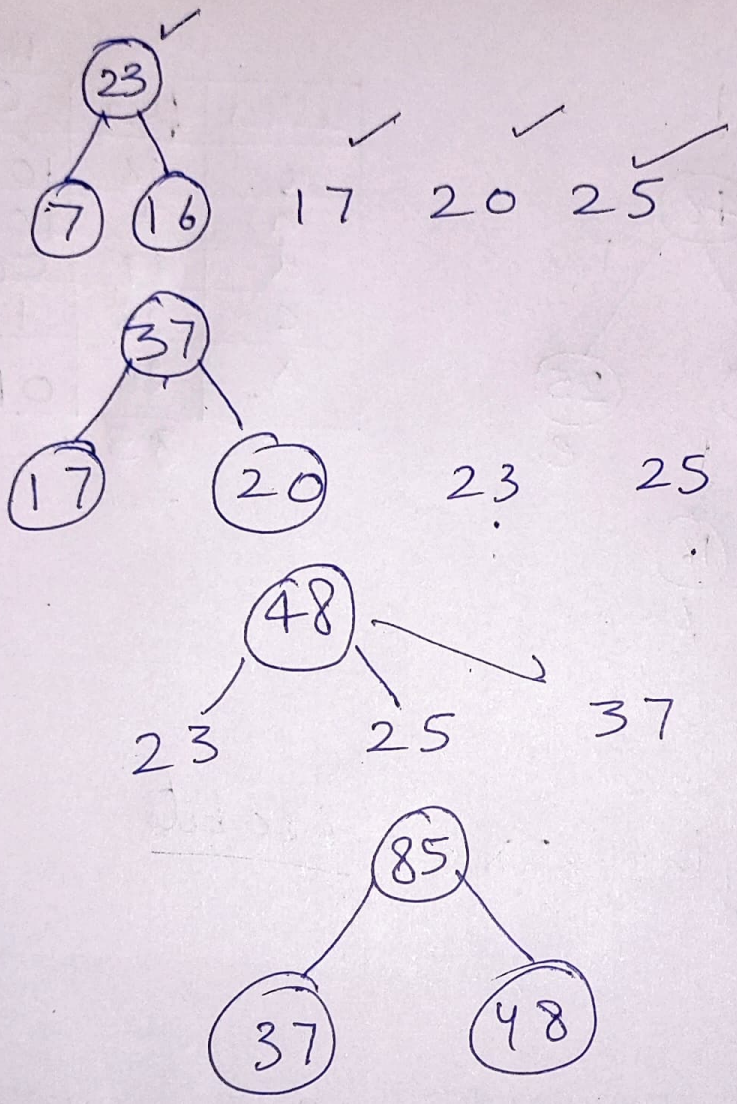
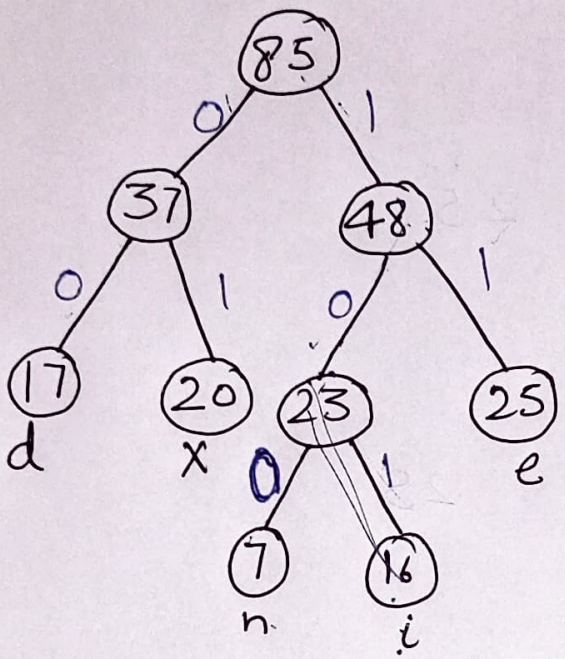
Q Let in a file the frequency of letters i, n, d, e, x are 16, 7, 17, 25, 20 respectively. find the Huffman's code of respective letters?





Letters	freq.	Huffman Code	fix code
i	16	101	000
n	7	100	001
d	17	00	010
e	25	11	011
x	20	01	100
	85		

ASCII

Message length = $85 \times 8 = 680$ bits

---

fix size

Message length = $\underbrace{85 \times 3}_{\text{Encode}} + \underbrace{8 \times 5 + 3 \times 5}_{\text{Table size}}$

$= 310$

Huffman

Message length = $\sum f_i d_i + (8 \times 5 + 2 \times 3 + 3 \times 2)$

$= 193 + 40 + 12$

$= 245$

Compression Ratio = 680/310 (fix)

= 680/245 (Huffman).

Knapsack Problem

Given: → n objects and a knapsack
→ Object i has weight w_i and the knapsack has a capacity m
→ If a fraction x_i, $0 \le x_i \le 1$, of object i is placed into the knapsack. The profit earned is $p_i x_i$
→ Knapsack size is m, total weight of all chosen objects to be most m.

Formal Representation:

$$\text{maximize} \sum_{1 \le i \le n} p_i x_i$$

$$\text{Subject to} \sum_{1 \le i \le n} w_i x_i \le m$$

$$\text{and} \quad 0 \le x_i \le 1, \quad 1 \le i \le n$$

Eg Consider the following instance of the knapsack problem:

$n = 3$ $\quad m = 20$

$(p_1, p_2, p_3) = (25, 24, 15)$ $\quad (x_1, x_2, x_3)$

$(w_1, w_2, w_3) = (18, 15, 10)$

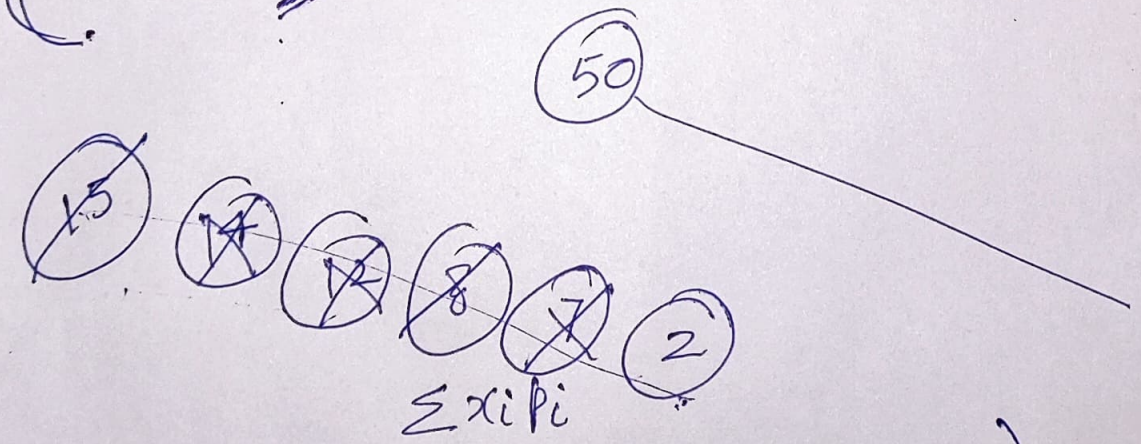
	(x_1, x_2, x_3)	$\sum w_i x_i$	$\sum p_i x_i$
1.	$(1/2, 1/3, 1/4)$	16.5	24.25
2.	$(1, 2/15, 0)$	20	28.2
3.	$(0, 2/3, 1)$	20	31
4.	$(0, 1, 1/2)$	20	31.5

Q

m=15

Objects (x)	Profit (P)	Weights (w)	P/w
1	10	2	5 ✓
2	5	3	1.6
3	15	5	3 ✓
4	7	7	1
5	6	1	6 ✓
6	18	4	4.5 ✓
7	3	1	3 ✓

$x(x_1$	x_2	x_3	x_4	x_5	x_6	$x_7)$
1	2/3	1	0	1	1	1



$\sum x_i p_i$

$(10 + 5 \times 2/3 + 15 + 0 + 6 + 18 + 3)$ ✓

→ P/w → decendy order

→

```
Algorithm Greedy_knapsack(m, n)
{
    U = m;
    for i = 1 to n
    {
        if (w[i] > U) then break;
        x[i] = 1.0; U = U - w[i];
    }
    if (i ≤ n) then x[i] = U/w[i];
}
```

Time Complexity = n (assume P/w already sorted)

$= n + n \log n$ (sorting)

$= n \log n$

Activity Selection Problem

Given

↳ Set of n activities $S = \{a_1, a_2 \text{ ---- } a_n\}$ that wish to access a common resource.

↳ Each activity a_i has starting time s_i and finishing time f_i where $0 \leq s_i \leq f_i \leq \infty$

↳ Activity a_i will occupy resource for $\{s_i, f_i\}$ interval.

~~↳ Activities a_i will occupy resource~~

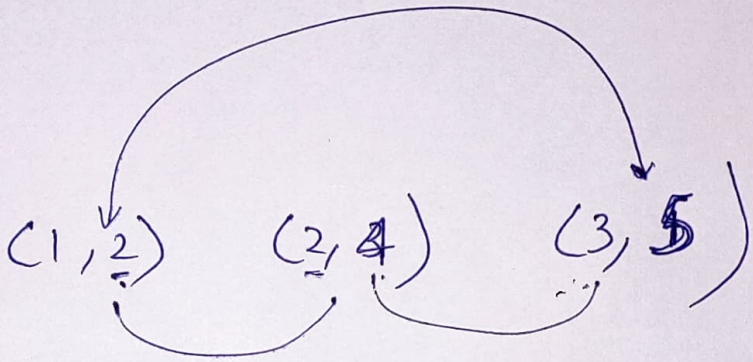
↳ Activities a_i & a_j are said to be compatible if the intervals $[s_i, f_i]$ & $[s_j, f_j]$ do not overlap

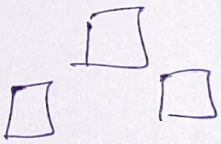
$$s_i \geq f_j \quad \text{or} \quad s_j \geq f_i$$

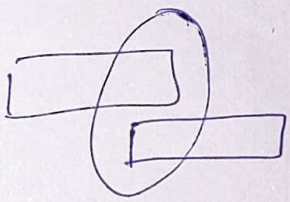
Maximum size subset of mutually compatible Activities.

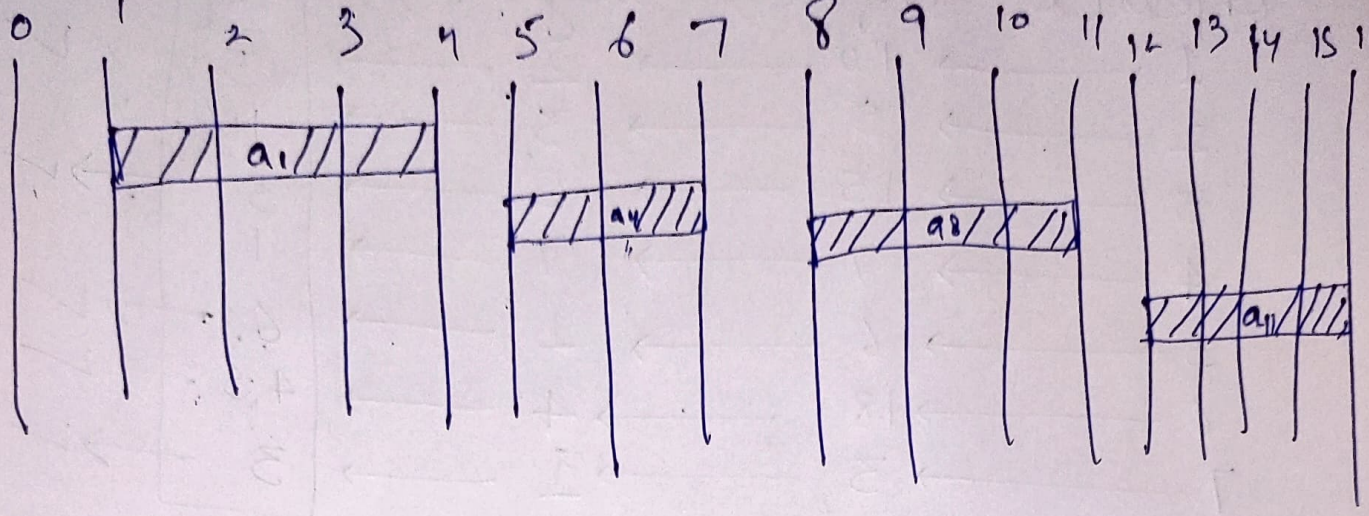
Eg

i	1	2	3	4	5	6	7	8	9	10	11	
s_i	1	3	0	5	3	5	6	8	8	2	12	
f_i	4	5	6	7	9	9	10	11	12	14	16	✓
	✓	✗	✗	✓	✗	✗	✗	✓	✗	✗	✓	









Sort acc. to finish time

```
Algorithm greedy-AS (s, f)
{
    n = s.length;
    A = {a1};
    K = 1;
    for m = 2 to n {
        if (s[m] ≥ f[K]) {
            A = 'A ∪ {am};
            K = m;
        }
    }
    return A;
}.
```

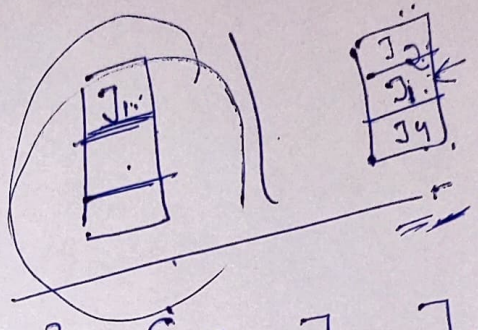
Complexity

Job sequencing with deadlines

Jobs	J_1	J_2	J_3	J_4	J_5
Profits	20	15	10	5	1
Deadlines	2	2	1	3	3

[Each job takes 1 unit of time]

Maximize Profit



0-1 J_2
1-2 J_1
2-3 J_4

Possible sequences: $\{J_2, J_1, J_4\}$ $\{J_1, J_2, J_4\}$.

Total Profit

Job Considered	slot assign	solution	Profit
			0
J_1	{1,2}	J_1	20
J_2	{0,1} {1,2}	J_1 J_2	20+15
J_3	{0,1} {1,2}	J_1 J_2	20+15
J_4	{0,1} {1,2} {2,3}	J_1 J_2 J_4	20+15+5
J_5	11	11	11